

R2S-EGO: Dual-Proxy Refinement for Sparse-Capture Real-to-Sim

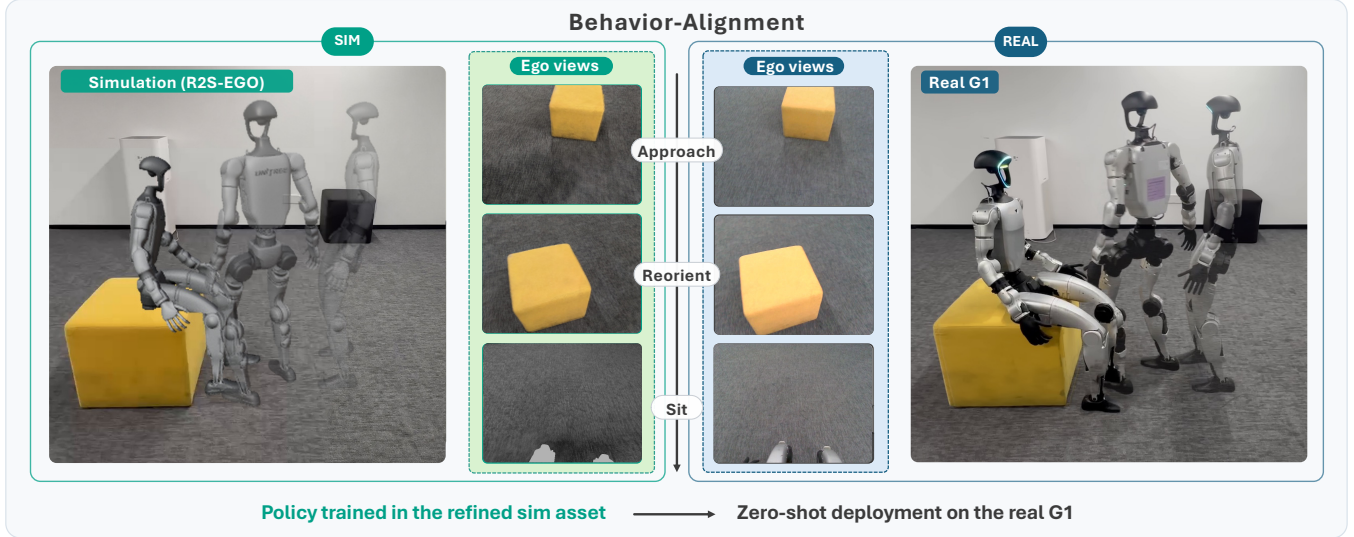


Fig. 1. Behavior-aligned third- and first-person sitting rollouts in simulation and on the real Unitree G1 humanoid. Teal panels show the policy rollout in the R2S-EGO simulation scene; navy panels show the corresponding real-robot rollout, aligned at approach, reorientation, and sitting. The center columns compare the simulated ego view with the onboard head-camera view at each phase, showing closely matched appearance and behavior across the transfer.

Abstract—Real-to-sim (R2S) depends on scene representations that render observations along robot ego trajectories. Dense multi-view capture limits per-environment capture-count efficiency, while sparse human capture can leave behavior-scoped robot views under-supported. Camera-controlled synthesis can fill these missing views, but its use in R2S requires behavior-admissible queries and capture-anchored structural conditioning. R2S-EGO couples a simulator-derived robot proxy, which defines a behavior-scoped executable query domain, with a capture-anchored geometry proxy that supplies structural conditions. Under a fixed budget, it selects current support deficits for which geometry is available. Generated observations are registered and assimilated at lower weight while real captures remain anchors; the geometry proxy also refreshes the collision surface, with dynamics and control fixed. On 48 frozen Unitree G1 ego views, rendered from three Replica scenes with controlled ground truth, six-view R2S-EGO reaches 19.062 dB PSNR versus 14.226 dB for the strongest reported R2S baseline. In one controlled real-G1 sitting protocol with five paired policy-training seeds and GaussGym as the baseline, success is $82.5\% \pm 6.8\%$ versus $10.0\% \pm 10.5\%$.

I. INTRODUCTION

Real-to-sim (R2S) turns real environments into repeatable robot-training scenes [1]–[3], commonly reconstructed from calibrated multi-view observations with neural or Gaussian representations [3]–[6]. Dense capture, however, limits per-environment capture-count efficiency, while human-convenient sparse capture and behavior-scoped robot cameras need not share poses, visible surfaces, or occlusions. The resulting under-support at robot views is the *capture-consumption support gap*.

Camera-controlled synthesis can add observations [7]–[10], but R2S requires both behavior-admissible queries and structural evidence tied to the captured scene. R2S-EGO couples two proxies (Fig. 2). A simulator-derived *robot proxy* maps fixed-controller rollouts through the camera mount to a behavior-scoped executable query domain; within it, fixed-budget selection targets current visual deficits having usable geometry. A *geometry proxy* aligns shape-prior regions to capture fragments and fuses them with NKSr-completed regions [11], yielding a capture-anchored structural hypothesis. Rendered at the same mounted-camera poses, it conditions generation on layout, visibility, depth order, and occlusion. VGGT registers retained frames to fixed real references; the frames then update the visual asset at lower weight while captures remain anchors. The fused geometry refreshes the collision surface between rounds, while robot dynamics and control remain fixed.

Across 48 frozen Unitree G1 ego views in three Replica scenes, six-view R2S-EGO reaches 19.062 dB PSNR versus 14.226 dB for the strongest reported R2S baseline. Across five independently trained policies per condition, controlled real-G1 sitting succeeds in 33/40 trials for R2S-EGO versus 4/40 for GaussGym; Fig. 1 shows the labeled sim/real phases and Fig. 6(a) a registered-pose comparison under its captioned limits. Our contributions are:

- We formulate sparse-capture R2S around support deficits within behavior-scoped executable robot views rather than uniform completion.
- We couple executable robot queries with capture-

anchored structural conditioning to select, register, and assimilate fixed-budget ego observations while refreshing the geometry/collision proxy.

- We evaluate frozen robot-view appearance and controlled real-G1 downstream transfer under explicit evidence boundaries.

II. RELATED WORK

Reconstructed and generated environments for robot learning: RialTo and URDFormer recover interactive or articulated simulation assets [12], [13]; SplatSim, RoboGSim, GaussGym, Vid2Sim, Wanderland, and RobotArena[∞] connect reconstructed or video-derived environments to robot learning or evaluation [1]–[3], [6], [14], [15]. Generative models predict action-conditioned observations or policy-training views [16]–[19]. GRAIL uses generated 3D content and human-object video priors for G1 loco-manipulation [20]; its video supplies motion rather than observations of a captured scene. R2S-EGO instead refines a specific sparse-capture environment along behavior-scoped robot views, introducing neither a simulator nor a policy learner.

Sparse scene recovery and generative refinement:

Sparse recovery includes radiance fields, Gaussians, few-view priors, and pointmap reconstruction [4], [5], [21]–[24]; 3DFIRES, WorldMirror, and SAM 3D infer additional structure [25]–[27]. Video diffusion and camera control provide temporal and calibrated-view priors [7]–[10], [28], [29], and generated or restored views can return to NeRF or 3DGS optimization [30]–[37]. R2S-EGO does not claim novelty for this feedback. Its distinction is an R2S interface in which a fixed embodiment, mounted camera, and behavior scope define queries, while capture-anchored structure constrains the observations that update the visual asset and geometry/collision proxy.

Active and robot-targeted view allocation: Next-best-view and active mapping acquire sensing poses for reconstruction [38]–[40]. Recent systems use Gaussian information gain, uncertainty, language, or imagined geometry [41]–[43]. These methods acquire real measurements for global reconstruction; R2S-EGO allocates synthetic observations inside an executable robot-query domain to current deficits having capture-anchored structural support.

III. METHOD

A. Problem Setting and Output

Let $\mathcal{C} = \{(I_i, \pi_i)\}_{i=1}^N$ be sparse calibrated RGB captures. A fixed registration maps them and initial asset $\mathcal{A}_0$ to simulator frame $\mathcal{W}$, yielding $\mathcal{C}^{\mathcal{W}}$ and $\mathcal{A}^0$. Simulator $\mathcal{S}$ supplies embodiment $\mathcal{E}$, dynamics, camera mount ${}^B T_C$, and control; behavior scope $\mathcal{B}$ specifies admissible configurations and motion. Without new real observations, R2S-EGO returns visual asset $\mathcal{A}^*$ and the collision surface derived from fused geometry. The robot proxy defines executable queries and current visual deficits; the geometry proxy supplies capture-anchored conditions, and a fixed budget selects eligible queries before synthesis. Registered outputs update the asset,

while NKSR geometry and collision are refreshed between rounds; dynamics and control stay fixed. Here l indexes the round and $\mathcal{A}^l$ its starting asset.

B. Dual-Proxy Ego-View Construction

1) *Capture-Anchored Geometry Proxy:* Calibrated multi-view tracks are triangulated into fragmentary geometry $\mathcal{M}_{\text{obs}} = (\mathcal{P}_{\text{obs}}, A_{\text{obs}})$, where the support record A_{obs} identifies contributing captures. A fixed vocabulary prompts SAM 3 [44], and each detected mask is passed automatically to SAM 3D [27] to obtain $\mathcal{P}_{\text{prior}} = \{\mathcal{P}_{\text{prior}}^m\}_m$. The proxy is then composed as

$$\mathcal{M}_{\text{scaf}}^l = \text{Compose}\left(\{T_m^*(\mathcal{P}_{\text{prior}}^m)\}_m, \text{NKSR}(\mathcal{P}_{\text{obs}}^{\neg\text{prior}} \cup \mathcal{P}(\mathcal{A}^l)^{\neg\text{prior}})\right). \quad (1)$$

$T_m^*(p) = s_m^* R_m^* p + t_m^*$ is a world-frame similarity estimated from real-prior overlap and multi-view silhouette consistency, then held fixed. Compose retains each aligned SAM 3D surface inside its object region and unions those surfaces with the NKSR reconstruction of the residual real fragments and current backend geometry. Rendering $\mathcal{M}_{\text{scaf}}^l$ supplies frame-aligned layout, visibility, and occlusion. Capture support anchors placement, but prior-only surfaces remain structural hypotheses rather than verified geometry.

2) *Behavior-Scoped Robot Proxy:* Each round reruns fixed behavior controller $\mu_{\mathcal{B}}$ with resampled initial conditions or references until 12 rollouts succeed. Forward kinematics and the fixed mount give candidate stream $\mathcal{V}_0^l = \{\mathcal{W} T_{C,t}\}_t$. Selection scores current deficits with usable geometry and retains temporal maxima:

$$\begin{aligned} \mathcal{W} T_{C,t} &= \mathcal{W} T_B(q_t) {}^B T_C, \\ s_l(t) &= u_l(t) g_l(t), \\ \mathcal{T}_l &= \text{TopK}_K(\text{NMS}_t\{(t, s_l(t))\}_t), \\ \mathcal{V}^l &= \{\mathcal{W} T_{C,t} : t \in \mathcal{T}_l\}, \quad |\mathcal{V}^l| = K. \end{aligned} \quad (2)$$

$u_l(t)$ is the fraction of pixels whose current-asset alpha/coverage lies below the 0.8 asset threshold, and $g_l(t)$ is the fraction whose proxy render passes a separate 0.8 geometry-validity threshold. The thresholds form pixel masks, not pose-level gates. Their product is a separable pose-ranking heuristic over the two marginal fractions, not a pixel-wise intersection score, and makes no independence claim. It is monotone in each scalar and low when either is small, so a pose strong in only one factor is demoted. Temporal NMS retains local maxima and TopK_K returns their K time indices. Thus $\mathcal{V}_0^l$ is the executable behavior-scoped domain and $\mathcal{V}^l$ its budgeted pose subset. These fixed-controller queries are reached by the mounted camera and do not represent learned-policy occupancy.

3) *Joint Ego-View Generation:* For target v , real capture $I_{i(v)}$ provides appearance and τ_v the camera path; a shared view index binds motion, rendered structure, and output

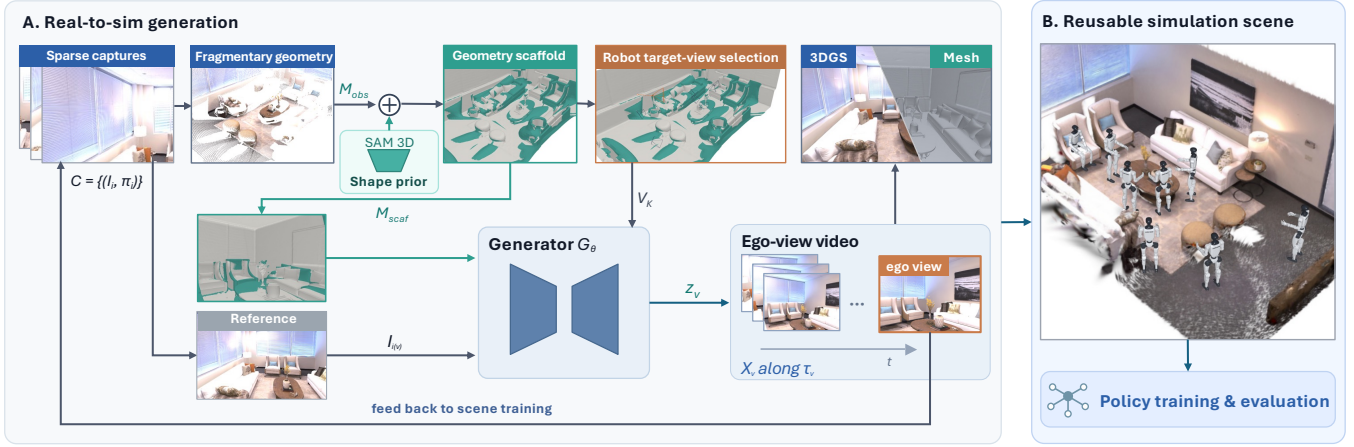


Fig. 2. **R2S-EGO pipeline.** R2S-EGO refines an existing simulation scene along behavior-scoped robot views. A simulator-derived robot proxy represents executable camera queries and identifies which remain under-supported, while a capture-anchored geometry proxy supplies view-aligned structural conditions. Registered ego observations update the visual asset as lower-weight pseudo-observations while real captures remain anchors; the refreshed geometry/collision proxy supports the next-round rollout and downstream simulation. All panels show the same `room_0` scene reconstructed from the same six captures. In the geometry panels, gray marks capture-observed surfaces and teal marks surfaces supplied by the shape prior.

registration:

$$\begin{aligned}
 z_v^l &= \mathcal{R}(\mathcal{M}_{scaf}^l, \tau_v), \\
 X_v^l &= G_\theta(I_{i(v)}, z_v^l, \bar{\tau}_v), \\
 \mathcal{D}_{ego}^l &= \{(X_v^{l, N_f}, \hat{\pi}_v^l) : v \in \mathcal{V}^l\}.
 \end{aligned} \tag{3}$$

N_f is the fixed sequence length: τ_v has N_f poses, z_v^l their rendered conditions, and $\bar{\tau}_v = \{\bar{\pi}_{v,t}\}_{t=1}^{N_f}$ their reference-relative poses (Sec. III-C). Only the terminal frame is retained. Query v controls generation but is not used as its training pose; VGGT estimates the frame pose and intrinsics relative to the fixed real reference, whose registered world pose is never replaced.

C. Scene Refinement and Implementation

Let $\bar{\mathcal{D}}_{ego}^l = \bigcup_{j=0}^l \mathcal{D}_{ego}^j$ collect registered generated observations through round l . Updating $\mathcal{A}^l$ uses this cumulative set at $w_{syn} = 0.5$ and registered captures at $w_{real} = 1$, so real data remain coordinate and appearance anchors. The output keeps the asset interface and installs current fused geometry as the collision surface; downstream dynamics and control are unchanged. Algorithm 1 gives the fixed-round execution.

Registration and geometry: Fixed $\mathcal{W}_{T_{cap}}$ maps captures to $\mathcal{W}$, giving $\mathcal{W}_{T_{C,i}} = \mathcal{W}_{T_{cap}}\pi_i$ and $\mathcal{A}^0 = \text{RegisterAsset}(\mathcal{A}_0, \mathcal{W}_{T_{cap}})$; $\mathcal{A}_{TB}$ maps frame B to frame A . Ground correction aligns z , places the origin at the first real capture, and calibrates scale separately; the transform applies to all assets, robot poses, and collision geometry. Each SAM 3 mask is passed automatically to SAM 3D [27] with public defaults and no manual instance choice. After excluding prior-covered objects, NKSR reconstructs remaining real points with current backend geometry and refreshes collision each round. Prior regions remain hypotheses with no independent metric/collision-accuracy evaluation.

Reference-conditioned generation: For each target, a fixed rule caches real reference $i(v)$ with greatest capture-anchored proxy overlap: the fraction of target-visible proxy

Algorithm 1 R2S-EGO Dual-Proxy Scene Refinement

Require: Captures $\mathcal{C}$; capture registration $\mathcal{W}_{T_{cap}}$; simulator $\mathcal{S}$; behavior scope $\mathcal{B}$; rounds L ; target-view budget K ; initial asset $\mathcal{A}_0$; fixed configuration $\mathcal{H}$

- 1: $(\mathcal{C}^{\mathcal{W}}, \mathcal{A}^0) \leftarrow \text{REGISTERINPUTS}(\mathcal{C}, \mathcal{A}_0, \mathcal{W}_{T_{cap}})$
- 2: $(\mathcal{M}_{obs}, \mathcal{P}_{prior}) \leftarrow \text{OBSERVEDANDPRIORGEOMETRY}(\mathcal{C}^{\mathcal{W}}; \mathcal{H})$
- 3: $\{T_m^*\}_m \leftarrow \text{ALIGNPRIOR}(\mathcal{M}_{obs}, \mathcal{P}_{prior}; \mathcal{H})$
 $\triangleright$ Prior transforms remain fixed across all rounds
- 4: $\bar{\mathcal{D}}_{ego}^{-1} \leftarrow \emptyset$
- 5: **for** $l = 0$ to $L - 1$ **do**
- 6: $\mathcal{M}_{scaf}^l \leftarrow \text{GEOMETRYPROXY}(\mathcal{M}_{obs}, \mathcal{P}_{prior}, \{T_m^*\}_m, \mathcal{A}^l; \mathcal{H})$
- 7: $\text{REPLACECOLLISIONMESH}(\mathcal{M}_{scaf}^l; \mathcal{H})$
- 8: $\mathcal{V}_0^l \leftarrow \text{SUCCESSFULEGPOSES}(\mathcal{S}, \mu_B, \mathcal{B}; \mathcal{H})$
 $\triangleright$ Resample failed rollouts until 12 succeed
- 9: $\{i(v)\}_v \leftarrow \text{SELECTREFERENCES}(\mathcal{C}^{\mathcal{W}}, \mathcal{V}_0^l, \mathcal{M}_{obs}; \mathcal{H})$
- 10: $\mathcal{V}^l \leftarrow \text{ROBOTPROXY}(\mathcal{V}_0^l, \mathcal{M}_{scaf}^l, \mathcal{A}^l, K; \mathcal{H})$
- 11: $\mathcal{D}_{ego}^l \leftarrow \text{EGOVIDEOREG}(\mathcal{C}^{\mathcal{W}}, \mathcal{M}_{scaf}^l, \mathcal{V}^l, \{i(v)\}_v; \mathcal{H})$
- 12: $\bar{\mathcal{D}}_{ego}^l \leftarrow \bar{\mathcal{D}}_{ego}^{l-1} \cup \mathcal{D}_{ego}^l$
- 13: $\mathcal{A}^{l+1} \leftarrow \text{UPDATEVISUALASSET}(\mathcal{A}^l, \mathcal{C}^{\mathcal{W}}, \bar{\mathcal{D}}_{ego}^l; \mathcal{H})$
- 14: **end for**
- 15: **return** refined scene: $\mathcal{A}^* \leftarrow \mathcal{A}^L$ and current collision mesh

points whose $\mathcal{A}_{obs}$ support is also visible in the candidate capture. Path τ_v interpolates to the target in N_f steps for ViewCrafter [10], with relative poses $\bar{\pi}_{v,t} = (\mathcal{W}_{T_{C,i(v)}})^{-1}\mathcal{W}_{T_{C,v,t}}$ and geometry rendered on the same path. Only the terminal generated frame is retained; VGGT registers it relative to the fixed reference before visual update rather than using the query pose directly.

Backends and controls: The visual asset is 3DGS. Fixed L , K , $N_f = 25$, generation, and reconstruction budgets are listed in Table I; other settings in $\mathcal{H}$ are chosen on a disjoint development split. Each round recomputes support, refreshes geometry/collision, and reruns controller rollouts, while the prior transform stays fixed. Frozen public checkpoints receive no scene-specific fine-tuning, and arms share registered inputs, preprocessing, and simulator settings. R2S-EGO uses 5k 3DGS steps per round. Its 36 clips generate $3 \times 12 \times 25 = 900$ video frames, of which only 36 terminal

TABLE I
Key implementation settings.

Setting	Configuration
Refinement rounds L	3
Successful rollouts per round	12
Views per round K	12
Frames per clip N_f	25; terminal retained
Synthetic weight w_{syn}	0.5; $w_{\text{real}} = 1$
Temporal NMS window	15
Asset alpha/coverage pixel threshold	0.8
Geometry validity pixel threshold	0.8
3DGS optimization	5k/round; 15k total
Generated-view calibration	VGGT

TABLE II

Quantitative comparison on frozen G1 ego views. Values average 48 views from three Replica scenes. R2S-EGO is the mean over five pre-specified pipeline seeds; other methods are single fixed runs. Vanilla 3DGS is the formal six-view Origin baseline; Diffix3D+ is the fixed official one-step postprocessor on GaussGym. GenFusion is a component-level visual-update comparison rather than an end-to-end R2S system and is evaluated under the matched six-view protocol. Best values are bold.

Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS-Alex $\downarrow$
Vanilla 3DGS	13.239	0.549	0.593
GenFusion [35]	14.012	0.481	0.571
Diffix3D+ [45]	14.059	0.461	0.552
GaussGym	14.226	0.567	0.628
R2S-EGO	19.062	0.757	0.273

frames are retained over $L = 3$; end-to-end construction time was not measured. The full method and its iterative and one-shot ablations use 15k total 3DGS steps, with generated and retained-frame budgets matched where applicable; recorded baseline budgets are method-specific and given below. Diffix3D+ is a fixed one-step postprocessor without an iterative budget. R2S-EGO is offline asset construction; total compute efficiency is not measured.

IV. EXPERIMENTS

We evaluate frozen ego-view appearance and capture-count efficiency, ablate the dual-proxy components, and compare matched sitting policies in simulation and under a controlled real-G1 protocol.

A. Setup

Benchmark protocol: We use three Replica scenes: *office_2*, *office_3*, and *room_0*. Each has six equal-interval inputs on a fixed trajectory and 16 G1-mounted targets: eight geometry-farthest and eight held-out routine-route placements. All 48 targets differ from inputs and retained pseudo-observation targets. The inputs, targets, and three qualitative views were frozen before comparisons, output inspection, hyperparameter choice, or ablation. Development and evaluation views are disjoint but share these scenes, behavior scope, and mounted-camera distribution; this tests unseen poses, not unseen behaviors, embodiments, or camera configurations. Replica Ptex supplies controlled GT; each

GT/render pair shares 640×480 intrinsics and camera-to-world matrix. Inputs and targets use Replica GT calibration, generated observations use VGGT, and physical captures also use VGGT when GT calibration is unavailable. Replica Ptex rendering supplies the controlled ground truth [46]. We report full-image PSNR, SSIM [47], and LPIPS-Alex [48] with the original 3DGS metrics, no mask, crop, alignment, or exposure/color correction; LPIPS uses the AlexNet backbone. Figure 6 and hardware trials provide complementary real-world evidence.

Comparison methods: All arms use the same six registered inputs, frozen public checkpoints, and no scene-specific fine-tuning. Vanilla 3DGS, GaussGym, Diffix3D+, and GenFusion are fixed runs; R2S-EGO and each ablation use five pre-specified paired seeds, averaging 48 views per seed then seeds. *Vanilla 3DGS* uses 10k-step per-scene optimization without generated video. R2S-EGO instead refines feed-forward asset $\mathcal{A}^0$ from a DUST3R/VGGT-family pointmap-to-Gaussian reconstructor [23], [49]; hence its *w/o ego-video* row differs from Vanilla despite common inputs. *GaussGym* uses its 30k sparse-stable SH3 condition; *Diffix3D+* [45] is its fixed official one-step postprocessor. *GenFusion* tests matched-six-view generation–reconstruction feedback, but lacks robot-scoped query allocation and a policy interface, so it appears only in the visual benchmark. R2S-EGO follows Algorithm 1. Per-seed dispersion is not reported for these appearance aggregates.

Downstream policy and hardware protocol: We follow GRAIL’s egocentric visual sim-to-real setup [20] and the SONIC stack [50], distilling a scene-aware sitting teacher into a student mapping head RGB and proprioception to SONIC latent tokens. Conditions share teacher, student, training, SONIC whole-body control (WBC), simulator physics, final R2S-EGO-derived scene collision mesh, and separately authored chair colliders; only the training visual asset changes, with no real-data fine-tuning. Training and evaluation vary robot/chair poses and chair backrest, height, shape, and appearance. Deployment receives no chair distance, reference ID, or manual reference initialization. Five paired seeds independently train each condition; within pair i , initialization, teacher data, and configuration match. Seeds, steps, and checkpoints are fixed before hardware outcomes. Each seed has 25 in-simulation episodes and eight real-G1 trials with manual resets approximating predefined relative-pose ranges. Methods alternate in a fixed order and the operator knows the condition. Success requires at least 3 s stable support with no intervention or load-bearing tether; the trained policy, not a trial, is the replicate. This proof of concept covers the declared sitting scope, not other behaviors or skills.

B. Ego-View Quality and View Efficiency

Same-view comparison: At fixed routine-route views, Fig. 3 shows baseline floaters, blur, or structural distortion at matched table, chair, and sofa regions, while R2S-EGO stays closer to GT. Over 48 views (Table II), R2S-EGO ranks first at 19.062 dB, 0.757 SSIM, and 0.273 LPIPS-Alex.

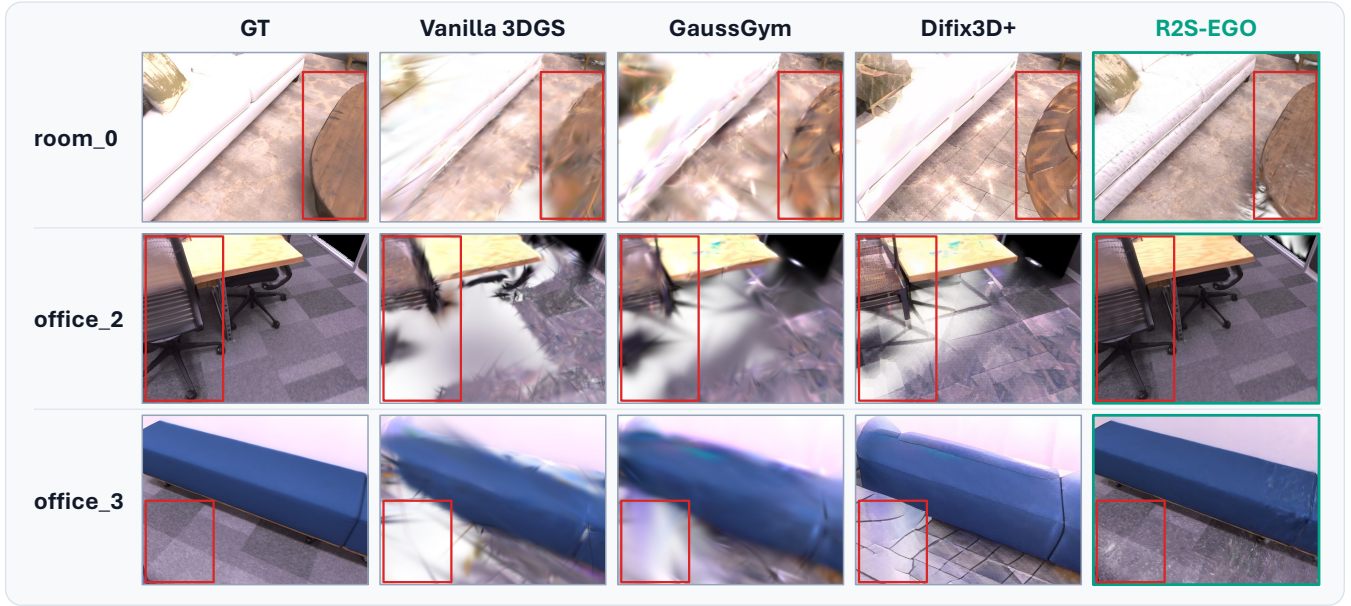


Fig. 3. **Same-camera qualitative comparison on frozen G1 ego views.** Each row uses the identical mounted-camera pose for GT, Vanilla 3DGS, GaussGym, Difix3D+, and R2S-EGO. The displayed routine-route view is fixed within each scene and is not chosen per method. Red boxes mark the same image regions in every column: the table rim, the office chair, and the sofa boundary, where Vanilla 3DGS and GaussGym degrade into floaters and blur and Difix3D+ distorts object structure, while R2S-EGO stays close to GT.

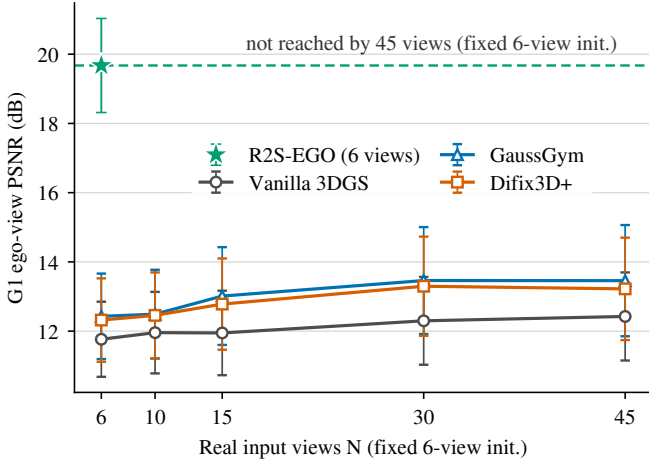


Fig. 4. **Measured real-image capture-count efficiency on G1 ego views under fixed six-view initialization.** Vanilla 3DGS, GaussGym, and Difix3D+ use nested input-image sets at the five displayed budgets; R2S-EGO remains a six-view point. Error bars are SEM over 16 fixed *office_2* target views, not seed uncertainty. No measured baseline setting through the 45-view endpoint reaches R2S-EGO@6. GenFusion is reserved for the matched six-view component comparison in Table II; because it is not an end-to-end R2S baseline, we do not construct a separate capture-count curve for it. Every baseline point retains the same six-view DUST3R initializer; this is a controlled-initialization input-count sweep, not end-to-end dense recapture reinitialized from all N inputs.

Baseline ordering is metric-dependent: GaussGym has the highest baseline PSNR/SSIM and Difix3D+ the lowest baseline LPIPS-Alex; GenFusion isolates generic generation–reconstruction feedback.

Controlled-initialization input-count sweep: In *office_2*, baselines use nested $N \in \{6, 10, 15, 30, 45\}$ sets, the same 16 targets, a fixed six-view DUST3R

initializer, and protocol-specific 10k-step Vanilla and 30k-step GaussGym optimization; Difix3D+ postprocesses GaussGym. Figure 4 reports PSNR, with target-view standard error of the mean (SEM) rather than seed uncertainty. R2S-EGO obtains 19.674 dB at six views; at 45, GaussGym, Difix3D+, and Vanilla obtain 13.458, 13.221, and 12.424 dB. No measured baseline reaches R2S-EGO@6. Because the initializer is fixed, this is the captioned controlled-initialization sweep, not end-to-end dense recapture. This evidence is limited to *office_2*, its 16 cameras, and PSNR: we infer neither an unmeasured catch-up count nor geometry equivalence. GenFusion has only its six-view result in Table II, not a curve. This is human input-count evidence, not a measurement of total compute efficiency.

C. R2S-EGO Component Ablation

Geometry-proxy visualization: Figure 5 pairs each RGB observation with its geometry-proxy rendering. Shared colors associate RGB regions with proxy objects within each pair, and the joint legend lists object IDs across both views. This visualization is qualitative and supplies no independent geometry-accuracy metric.

Row-wise ablation protocol: All variants use six inputs and 48 targets (Table III). *w/o SAM 3D grounding* removes the object prior, leaving NKSR for conditioning, collision, and later rollouts. *w/o ego-video* reduces to feed-forward $\mathcal{A}^0$. *w/o robot-proxy allocation* matches geometry, generated/retained-frame counts, and optimization but samples trajectories without robot feasibility or deficit/coverage ranking. The one-shot variant selects and retains 36 targets once, performs one 15k-step update, and omits intermediate

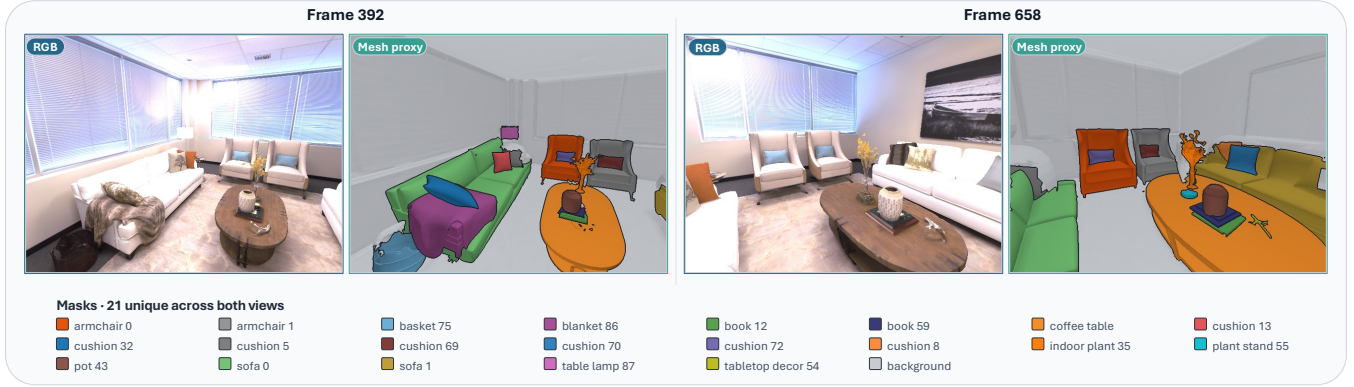


Fig. 5. **Two-view RGB-geometry-proxy correspondence.** Each card groups one `room_0` view (frames 392 and 658): the RGB observation followed by its geometry-proxy rendering. Shared mask colors denote object-level associations; the joint legend contains 21 object masks plus the background mesh. These annotations are qualitative and are not supplied to the generator.

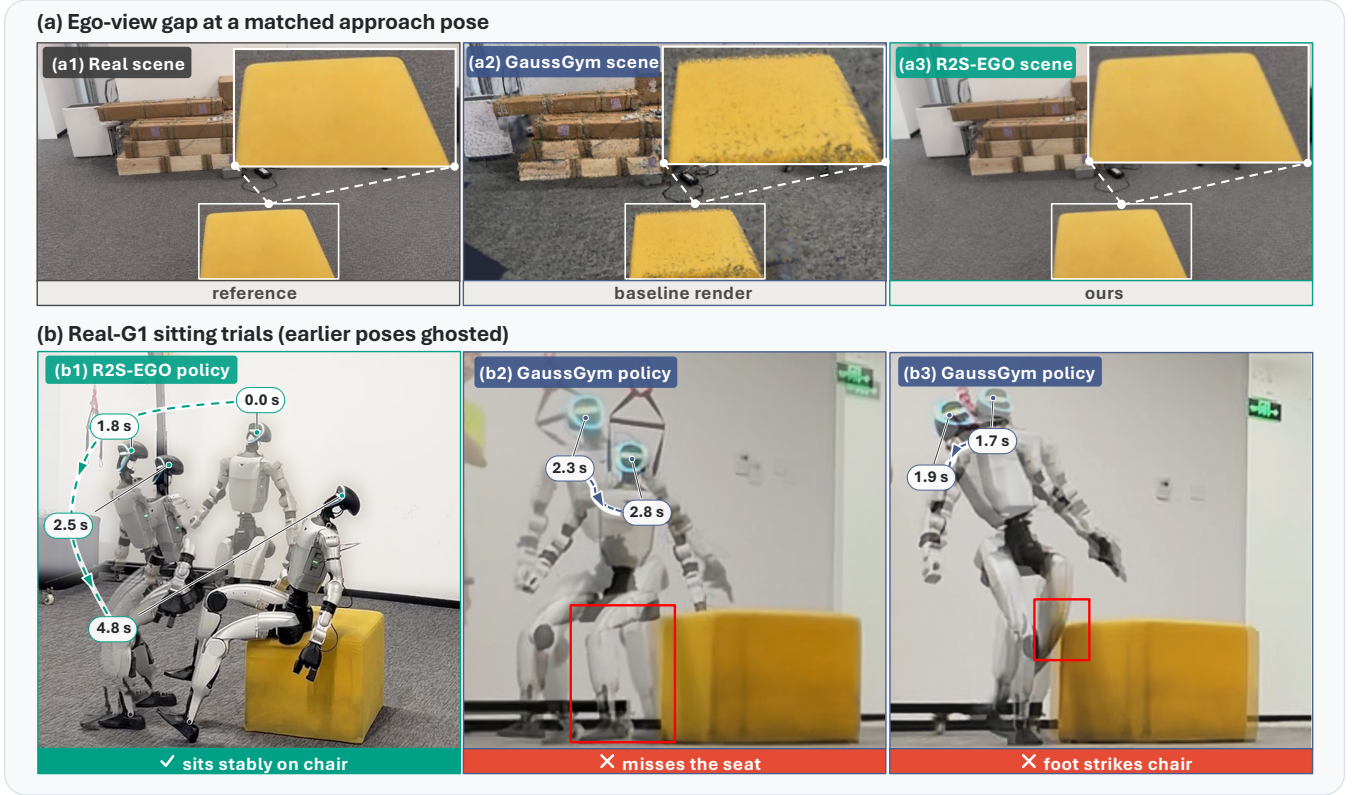


Fig. 6. **Registered-pose qualitative comparison and separate real-G1 sitting outcomes.** (a) The real approach image a1 was excluded from the 29-image reconstruction inputs but registered jointly for pose recovery and mapped by the recorded 29-camera Sim3 alignment; a1/a3 are pose-coincident and uncropped. Panel a2 is a genuine 6k-step early-stopped GaussGym qualitative render—distinct from the 30k Replica assets behind Table II—uniformly cropped to the real-camera FOV; its residual $\sim 9\%$ lateral discrepancy is attributed mainly to focal length, so the triplet is not a pixel-aligned overlay. The same chair region is boxed and enlarged; proximity to retained pseudo-observation targets is unverified. (b) Timestamped composites show an R2S-EGO success (b1) and two GaussGym failures whose causes are boxed in red (b2,b3); the separate five-policy protocol yields 33/40 versus 4/40. Because b1 spans the whole approach, its camera is farther from the chair than in b2/b3.

geometry/collision, rerollout, and reranking, matching total frames and optimization.

Component contributions: Without ego video, $\mathcal{A}^0$ loses 5.645 dB. Random trajectories reduce PSNR from 19.062 to 16.182 dB and SSIM from 0.757 to 0.661 while LPIPS-Alex increases from 0.273 to 0.406. Because this jointly changes candidate construction and ranking, the 2.880 dB difference

supports their coupled procedure, not individual effects. Removing SAM 3D gives 16.384 dB; the matched one-shot variant gives 17.508 versus 19.062 dB. SAM 3D removal is slightly higher than random allocation on PSNR/SSIM (16.384/0.671 versus 16.182/0.661) but lower on LPIPS-Alex (0.412 versus 0.406). These aggregate differences are neither significance nor per-scene-consistency claims; per-

TABLE III

Component ablation of R2S-EGO. Mean over five paired pipeline seeds with six real inputs and 48 frozen G1 views. Best values are bold.

Variant	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS-Alex $\downarrow$
w/o ego video ($\mathcal{A}^0$)	13.417	0.552	0.588
w/o robot-proxy allocation	16.182	0.661	0.406
w/o SAM 3D grounding	16.384	0.671	0.412
w/o iterative refinement	17.508	0.706	0.334
Full R2S-EGO	19.062	0.757	0.273

seed dispersion is not reported.

D. Real-Robot Sitting Validation

Across five independently trained policies, GaussGym and R2S-EGO obtain $87.2\% \pm 5.2\%$ and $97.6\% \pm 2.2\%$ in simulation, and $10.0\% \pm 10.5\%$ and $82.5\% \pm 6.8\%$ on the real G1 (Table IV); mean and sample standard deviation are across policies. The hardware ordering holds for every seed. Pooled 4/40 and 33/40 are descriptive because trials nest within policies; paired counts are (0, 0, 1, 2, 1) versus (7, 7, 7, 6, 6). Both conditions are high in simulation, whereas the separated hardware results support an association with the refined visual asset under this protocol. Figure 6(a) is a registered-pose qualitative comparison with the captioned focal-length, 6k early-stop, and distinct-asset limits; it is not evidence for the hardware association. Panel (b)’s annotated failures are illustrative, not a causal diagnosis. The slack tether bears no nominal load; intervention or load-bearing engagement is failure, and detachment follows a successful tethered sit. Accordingly, the illustrated GaussGym failures remain tethered and the R2S-EGO success is untethered.

TABLE IV

Success rates in simulation and real-world deployment. Mean $\pm$ sample standard deviation across five independently trained policies; 25 simulation episodes and 8 real-G1 trials per policy, spanning varied robot-chair poses and chair types. Best values are bold.

Training scene	Sim (% , mean $\pm$ std)	Real (% , mean $\pm$ std)
GaussGym	87.2 $\pm$ 5.2	10.0 $\pm$ 10.5
R2S-EGO	97.6 $\pm$ 2.2	82.5 $\pm$ 6.8

V. CONCLUSION

R2S-EGO couples executable robot-view queries and current-deficit selection with capture-anchored structural conditioning, then assimilates registered observations at lower weight and refreshes collision geometry. Across 48 frozen G1 views in three scenes, six-view R2S-EGO reaches 19.062 versus 14.226 dB; in controlled real-G1 sitting, R2S-EGO and GaussGym succeed in 33/40 and 4/40 trials, with the same ordering for every policy seed. These results support behavior-scoped sparse-capture refinement in the evaluated settings.

The full-reference appearance benchmark uses Replica-rendered inputs and ground truth, so its controlled metrics

do not directly measure physical-scene image quality. The input-count study retains a fixed six-view initializer and is not an end-to-end dense-recapture comparison; total compute efficiency and the query-to-VGGT registered-pose displacement are unmeasured. Prior-completed geometry remains a structural hypothesis, not scene truth; we measure neither independent metric/collision accuracy nor its contribution apart from visual refinement. The robot proxy covers a declared behavior scope, not learned-policy occupancy, and pseudo-observation updates can propagate appearance or structure errors. Hardware evidence is limited to five independently trained policies per visual asset, one controlled sitting protocol, and GaussGym as the comparison; manual resets approximate predefined relative-pose ranges and the operator knows the condition. Other behaviors, including manipulation and fine hand-object interaction, remain future work.

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